

Assignment 1

KAI-HONG SU

Table of contents

Introduction	1
Research Question section	1
Dataset Introduction section	1
Dataset Description subsection	2
Results section	4

Introduction

The dataset analyses customer behavior using a marketing dataset. In the dataset, there are variables for marketing spend, visits, purchase amount, and segment for each customer. With this information we can understand whether the marketing spend on customers has positive effects.

Research Question section

This report aims to know the relationship between marketing spend and purchase amount, and whether high marketing spend will cause more visits.

Dataset Introduction section

	customer_id	signup_date	marketing_spend	visits	purchase_amount	segment
1	35480211_c0001	2025-10-26	150.96	6	102.99	Consumer
2	35480211_c0002	Jan 31, 2026	27.83	6	51.72	Enterprise
3	35480211_c0003	2025-05-28	244.53	7	260.50	Consumer
4	35480211_c0004	2024-11-25	220.21	7	193.02	Consumer
5	35480211_c0005	2025-03-26	122.72	4	257.15	SMB
	notes					

1
 2 returned
 3 contacted
 4 N/A
 5 no response

	marketing_spend	purchase_amount	visits	segment
1	150.96	102.99	6	Consumer
2	27.83	51.72	6	Enterprise
3	244.53	260.50	7	Consumer
4	220.21	193.02	7	Consumer
5	122.72	257.15	4	SMB

This dataset contains customer information on marketing activities and purchasing behavior. It includes customer id, signup date, marketing spend, visits, purchase amount, segment, and notes. The dataset is used to analyse whether marketing spend has positive effects on customer behavior.

Before the analysis, a few cleaning steps were done. First, the report only needs marketing_spend, purchase_amount, segment, and visits, so the other columns were removed as they were not relevant to the analysis. Moreover, rows with NA values were removed to increase the quality of the analysis.

Dataset Description subsection

```
'data.frame':  132 obs. of  4 variables:
 $ marketing_spend: num  151 27.8 244.5 220.2 122.7 ...
 $ purchase_amount: num  103 51.7 260.5 193 257.1 ...
 $ visits         : int   6 6 7 7 4 7 2 6 5 0 ...
 $ segment        : chr  "Consumer" "Enterprise" "Consumer" "Consumer" ...
```

The dataset contains 132 observations and 4 variables. The four variables used in the analysis are marketing_spend, purchase_amount, segment, and visits. Only segment is a character variable, and the remaining variables are numeric, with marketing_spend and purchase_amount stored as doubles and visits stored as integer.

Table 1: Summarise of the data

avg_marketing	avg_purchase	avg_visits
152.0372	157.6592	5.795454

Table 1 summarises the key statistics of the dataset. The average marketing spend is around 152.04, while the average purchase amount is around 157.66. Moreover, average number of visits is 5.8.

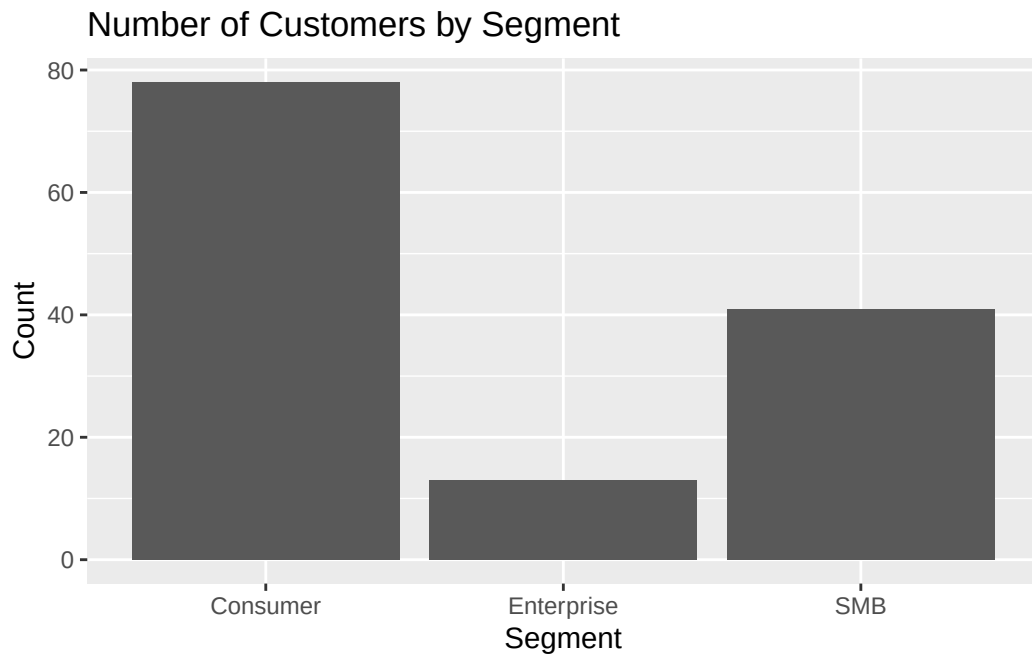


Figure 1: Bar chart show number of customers in different segment

Figure 1 shows the number of customers in each segment. From this chart, we can see that most customers in the dataset are in the Consumer segment.

Results section

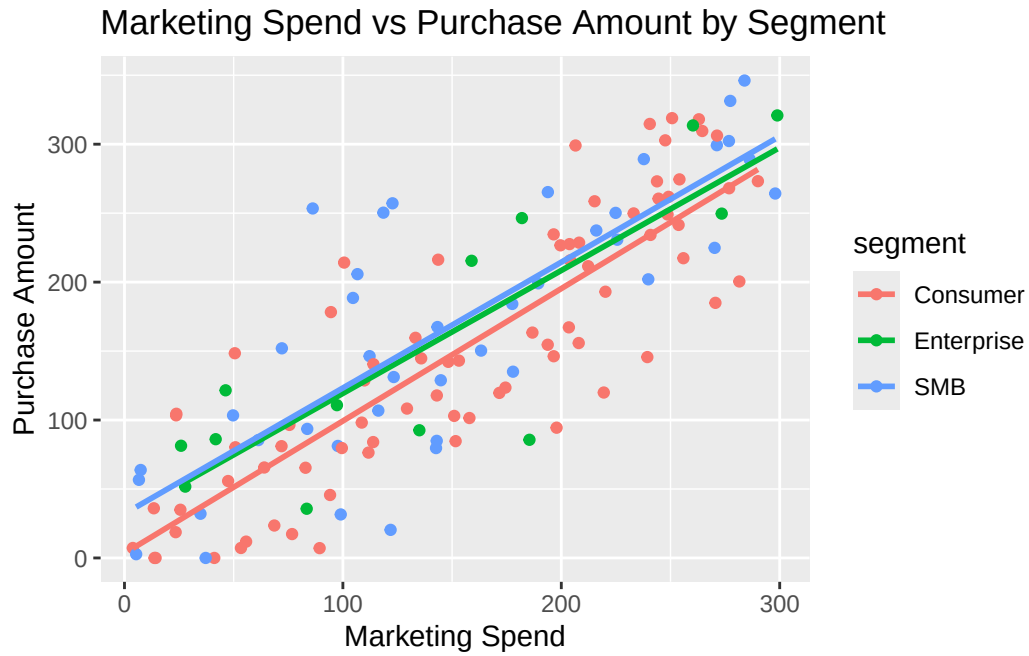


Figure 2: Relationship between marketing spend and purchase amount in different segments

Figure 2 shows a positive relationship between marketing spend and purchase amount across all three segments. The correlation between marketing spend and purchase amount is 0.83, indicating a strong positive association. As marketing spend increases, purchase amount also tends to increase for Consumer, Enterprise, and SMB customers.

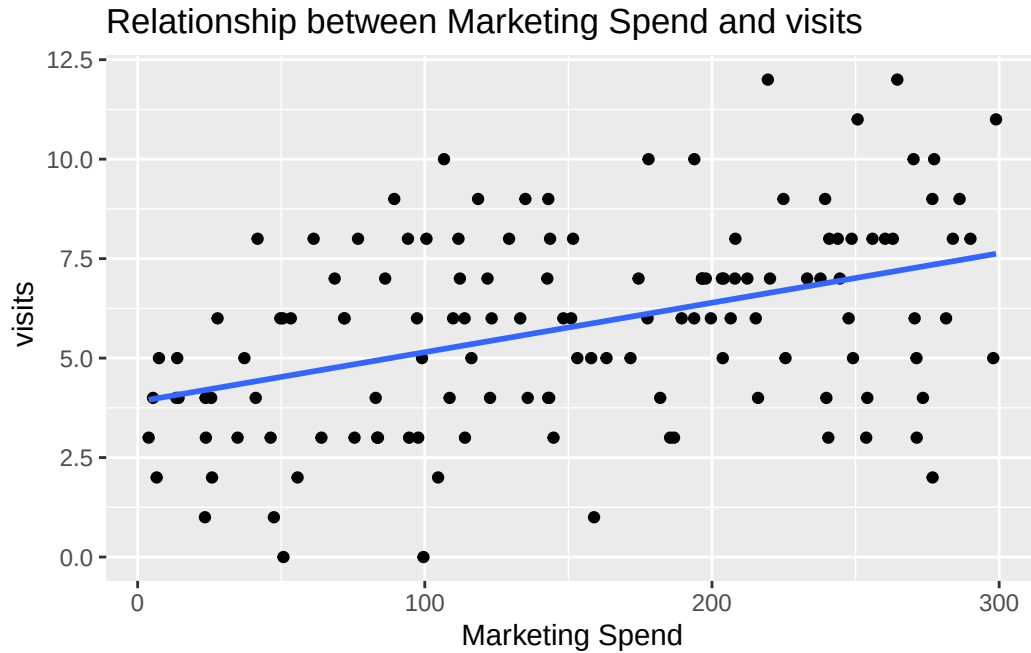


Figure 3: Relationship between marketing spend and visits

Figure 3 shows a positive relationship between marketing spend and visits. The correlation between marketing spend and visits is 0.42, indicating a positive but weak association. To sum up, customers with higher marketing spend tend to have more visits. However, the wide spread of the points suggest that there may be other factors affecting customer visits.